

1. Create a comparison table listing RAID 0, RAID 1, RAID 5, and RAID 10, including columns for minimum number of disks, data redundancy, speed, and a real-world use case for each (such as Instagram photo storage, YouTube video servers, etc.).

RAID Level	Minimum Disk	Data Redundancy	Speed	Real-world Use Case
RAID 0	2	None – data is striped across disks	Very high	Temporary video-editing files or gaming data where speed is more important than data safety.
RAID 1	2	Yes – data is mirrored on both disks	Good	Small business server storing important documents or user files.
RAID 5	3	Yes – distributed parity protects against 1 disk failure	Good reads / Moderate writes	YouTube-style video storage servers or file servers where large amount of data need reasonable protection.
RAID 10	4	Yes – combines mirroring and striping	Very high	Instagram-style photo storage, databases or high-traffic

				application servers.
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2. Explain with your own example how RAID 1 provides data redundancy, using a scenario like storing WhatsApp chat backups on two disks, and describe what happens if one disk fails.

RAID 1 Data Redundancy Example

- Imagine there are two 500 GB disks in your computer and configure them as a RAID 1.
- Suppose, you store your WhatsApp chat Backup as: WhatsApp_Backup.zip – 10 GB.
- With RAID 1, the same data is mirrored onto the both disks.
- Now, Disk 1 will have WhatsApp backup of 10 GB and Disk 2 will have WhatsApp backup of 10 GB.
- So, there are two identical copies of the backup.

What happens if Disk 1 fails?

- If Disk 1 suddenly stops working then the data is still available on Disk 2.
- You can continue accessing your WhatsApp backup without losing the stored data. You then replace the failed Disk 1 with a new disk and RAID 1 mirrors the data from Disk 2 onto the new disk.

3. Simulate a RAID 1 setup using VirtualBox and Ubuntu: create two virtual hard disks, use the 'mdadm' tool to configure them in RAID 1, and take a screenshot of the output from the command 'cat /proc/mdstat' showing the RAID array.

```
vboxuser@Meridian-Linux-Server:~$ cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sdc[1] sdb[0]
      5237760 blocks super 1.2 [2/2] [UU]
      bitmap: 0/1 pages [0KB], 65536KB chunk

unused devices: <none>
vboxuser@Meridian-Linux-Server:~$
```

4. Delete a file from your RAID 1 array, then power off one of the virtual disks (simulate a disk failure), and check if the file is still accessible from the RAID array.

```
vboxuser@Meridian-Linux-Server:~$ cat /proc/mdstat
Personalities : [raid1]
md0 : active raid1 sdb[0]
      5237760 blocks super 1.2 [2/1] [U_]
      bitmap: 0/1 pages [0KB], 65536KB chunk

unused devices: <none>
vboxuser@Meridian-Linux-Server:~$ cat /mnt/raid1/testfile.txt
RAID 1 test file
vboxuser@Meridian-Linux-Server:~$ _
```

5. Use ChatGPT to ask: 'What are the main differences between RAID 5 and RAID 10 for a cloud storage service like Google Drive?' Summarize the answer in 3 bullet points in your own words.
- RAID 5 – Uses at least 3 disks and stores parity information. So, it can survive one disk failure. It provides good storage efficiency but has slower write performance because of parity calculations.
 - RAID 10 – Uses at least 4 disks and combines mirroring and striping. It can provide faster read/write performance and better fault tolerance but it uses more storage space because data is duplicated.
 - For a cloud storage service like Google Drive – RAID 5 offers better storage efficiency while RAID 10 offers higher performance and redundancy. Large cloud providers may use more advanced storage systems rather than relying on a single RAID level.